

MPM Standardization Roadmap

Document Control Number: MPM-RMP-2026-003

Effective Date: October 2026

Associated Baseline Documents: MPM-CHR-2026-001 (Charter), MPM-GOV-2026-002 (Governance Structure)

Executive Summary

The MPM Standardization Roadmap defines the multi-year engineering trajectory, release schedules, and technical milestones for the Mechanical Power Module Consortium. The core objective is to transition distributed mechanical torque aggregation from custom industrial engineering to a globally standardized, vendor-neutral hardware primitive.

By establishing clear technical baselines across four distinct phases (2026–2030), this roadmap guides participating manufacturers, project developers, and certifying bodies toward interoperability for MPM units (77 kW to 1 MW class), Multi-MPM Aggregator Adapters (10 MW / 7.2 GWh/month blocks), and dedicated single-use off-grid systems.

1. Multi-Phase Roadmap Timeline (2026 – 2030)

2026	2027	2028	2029	2030
------	------	------	------	------

----- PHASE 1 -----	----- PHASE 2 -----	----- PHASE 3 -----	----- PHASE 4 -----
---------------------	---------------------	---------------------	---------------------

Physical & Spline Baseline Multi-Input Aggregation &	Single-Use Industrial	Utility Stacking
--	-----------------------	------------------

(77 kW Core Specs)	& Thermal Loops	Direct Coupling (1 MW)	Global Certification
--------------------	-----------------	------------------------	----------------------

Phase 1: Physical Interfaces & Spline Tolerances (Q4 2026 – Q2 2027)

Primary Lead: TWG-1 (Mechanical & Harmonics)

Focuses on establishing the baseline physical and mechanical boundary standards for individual 77 kW Mechanical Power Modules.

Core Deliverables:

MPM-SPEC-101 (Drive Spline & Coupling Geometry): Dimensional standardization of lower-pulley male/female drive splines, engagement keyways, and rapid-locking mechanism tolerances.

MPM-SPEC-102 (Container Chassis Lock Points): ISO-compliant corner casting geometries, anchor points, and structural load ratings for standard module enclosures.

MPM-SPEC-103 (Dynamic Balance & Baseline Harmonics): Maximum allowable rotational unbalance limits ($G2.5$ balance grade per ISO 1940-1) for individual MPM shaft assemblies operating up to $3,600\text{ RPM}$.

Phase 2: Multi-Input Aggregation & Auxiliary Thermal Loops (Q3 2027 – Q1 2028)

Primary Lead: TWG-1 & TWG-2 (Thermal & Fluid Loops)

Focuses on the containerized Multi-MPM Adapter framework, enabling multi-shaft torque consolidation while isolating harmful torsional vibrations.

Core Deliverables:

MPM-SPEC-201 (Multi-MPM Adapter Mechanical Intake): Gear train interfaces, clutch disengagement protocols for isolated unit servicing, and master gear reduction ratios for 10-MPM and 14-MPM arrays.

MPM-SPEC-202 (Torsional Resonance & Vibration Dampening): Guidelines for fluid coupling integration, elastomeric dampening hubs, and active anti-harmonic tuning to prevent destructive gear tooth wear under uneven multi-engine input.

MPM-SPEC-203 (Auxiliary Cooling & Quick-Disconnect Flanges): Pressurized lubrication manifolds, quick-disconnect fluid fittings ($ISO\ 16028$ compliant), and brine/seawater heat exchanger integration for 100% continuous duty thermal dissipation (1 MW thermal rejection threshold).

Phase 3: Single-Use Industrial Direct Coupling & Scale-Up (Q2 2028 – Q4 2028)

Primary Lead: TWG-3 (Single-Use & Off-Grid Integration)

Extends standard torque output interfaces directly into high-value, off-grid industrial machinery, bypassing grid transmission queues and power conversion losses.

Core Deliverables:

MPM-SPEC-301 (Megawatt-Scale Electrolyzer Direct Drive): Mechanical-to-fluid power interface standards for dedicated hydrogen production arrays, calibrated for 10 MW input blocks (7.2 GWh/month nominal equivalent).

MPM-SPEC-302 (High-Pressure Desalination & Uplift Pumping): Direct mechanical coupling specs for Reverse Osmosis (RO) positive displacement pumps and high-head mountain water transfer pumps.

MPM-SPEC-303 (Industrial Airflow Drive for Direct Air Capture): Direct torque transmission guidelines for megawatt-scale atmospheric CO_2 capture intake fans.

MPM-SPEC-304 (1 MW Class Single-Unit MPM Specification): Upscaled individual module physical footprint, shaft torque envelope, and spline ratings for megawatt-class single prime movers.

Phase 4: Utility Integration, 3-High Stacking & Global Interoperability (Q1 2029 – Q4 2030)

Primary Lead: TWG-4 (Grid, Safety & Regulatory) & CCB

Enables hyperscale deployment, vertical array stacking, utility grid interconnection, and global certification compliance.

Core Deliverables:

MPM-SPEC-401 (Multi-Tier Physical Stacking Mechanics): Structural interlocking, dynamic resonance suppression, and load-bearing framing for 3-high containerized adapter arrays ($30 \text{ MPMs} \rightarrow 30 \text{ MW}$ total, generating up to 21.6 GWh/month).

MPM-SPEC-402 (Synchronous Generator Matching & Grid Interconnection): Standardized automatic synchronizer interfaces, grid protection relay logic, multi-input emergency shutdown (ESD) loops, and anti-islanding safety protocols.

MPM-SPEC-403 (Carbon Accounting & Operational Telemetry Protocols): Encryption standards and data format schemas for automated, tamper-proof reporting of mechanical work output to international carbon credit registries.

MPM-CERT-001 (Consortium Interoperability Certification Manual): Test bench procedures, field audit criteria, and benchmark suites executed by accredited test facilities for granting the Consortium Interoperability Seal.

2. Technical Working Group Deliverables Matrix

Standard Code (RCS)	Working Group	Standard Title	Target Stage 2 (CS)	Target Stage 3
MPM-SPEC-101	TWG-1	Drive Spline & Coupling Geometry	Q1 2027	Q2 2027
MPM-SPEC-102	TWG-1	Container Chassis Lock Points	Q1 2027	Q2 2027
MPM-SPEC-103	TWG-1	Dynamic Balance & Harmonic Limits	Q2 2027	Q3 2027
MPM-SPEC-201 2027 Q1 2028	TWG-1 / TWG-2	Multi-MPM Adapter Mechanical Intake		Q4 2027
MPM-SPEC-202	TWG-1	Torsional Resonance & Isolation	Q4 2027	Q1 2028
MPM-SPEC-203	TWG-2	Brine/Seawater Cooling Interfaces	Q1 2028	Q2 2028
MPM-SPEC-301 Q4 2028	TWG-3	Electrolyzer Direct Coupling (\$10\text{ MW}\$)		Q3 2028
MPM-SPEC-302 Q4 2028	TWG-3	High-Pressure RO & Uplift Pump Interfaces		Q3 2028
MPM-SPEC-304 Q4 2028	TWG-1 / TWG-3	\$1\text{ MW}\$ Class MPM Module Standard		Q1 2029
MPM-SPEC-401 2029 Q3 2029	TWG-1 / TWG-4	3-High Array Stacking (\$30\text{ MW}\$)		Q2 2029
MPM-SPEC-402 2029	TWG-4	Utility Grid Relaying & Generator Matching	Q3 2029	Q4 2029

3. Engineering Benchmarks & Key Metrics

To ensure seamless field interoperability across diverse vendor hardware, all candidate specifications must validate performance against fixed mathematical thresholds:

3.1 Shaft Torque & Power Density Benchmarks

3.2 77 kW Baseline Module:

$$T = \frac{P}{\omega} \approx \frac{77\text{ kW}}{1,800\text{ RPM}} \approx 408.5\text{ N}\cdot\text{m}$$

1 MW Upscaled Module:

$$T = \frac{1,000\text{ kW}}{1,800\text{ RPM}} \approx 5,305\text{ N}\cdot\text{m}$$

3.3 Monthly Energy Output Capacity

3.4 10 MW Adapter Array Baseline:

$$P_{\text{monthly}} = 10\text{ MW} \times 24\text{ hours/day} \times 30\text{ days} = 7,200\text{ MWh} = 7.2\text{ GWh/month}$$

30 MW (3-High Stacked Array):

$$P_{\text{monthly}} = 30\text{ MW} \times 24\text{ hours/day} \times 30\text{ days} = 21,600\text{ MWh} = 21.6\text{ GWh/month}$$

3.3 Dynamic Balance & Vibration Suppression

Maximum permissible lateral movement at full rotational speed: $< 0.05\text{ mm/s}$ velocity RMS.

Torsional vibration transmission efficiency across multi-engine gearbox inputs: $\leq 2.0\%$ peak torque oscillation across 5 Hz to 500 Hz spectrum.

4. Governance Integration & Revision Cadence

Working Group Execution: Draft specifications originate in the designated TWG and move through the four-stage lifecycle detailed in the MPM Governance Structure (MPM-GOV-2026-002).

Annual Roadmap Review: In Q4 of each operational year, the Technical Steering Committee (TSC) shall publish an updated revision of this roadmap, incorporating market feedback, power density advancements, and emerging industrial single-use applications.

Open Standards Alignment: Core spline profiles and fluid interfaces developed under this roadmap remain subject to the Consortium's Mandatory Royalty-Free / FRAND Core Licensing Framework.